INCEPTION REPORT

Project Title - Efficiency and Bus Fleet Management: Strategic Communications and quantifying GHG and other co-benefits (#BBUS)

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1. Context

CEEW analysis shows that more than 230 million users depend on bus systems for their daily mobility requirements (Jha 2025). The majority of these users can be classified into a 'captive user group' having limited access to personal/private mobility modes. These populations comprise vulnerable road users such as women, the elderly, children and individuals from the economically weaker sections of society. Efficient public transport for these groups is not just about convenience, but a prerequisite for their socio-economic functioning. Indian cities are urbanising at a rapid pace, with around 600 million people expected to live in cities by 2036 (Kouamé 2024). This urban expansion is generating enormous travel demand, but public bus systems are not keeping pace with it. Historically, Indian cities were dependent on public transport, but in recent decades, the trends have reversed in favour of private vehicles (Dawda and Desai 2025). Moreover, the supply of public transport is inequitable, as more than 60% of India's bus transport fleet is concentrated in just 9 metro cities (Davidson and Ray 2025). Overall, public transport, especially bus systems, are facing challenges such as limited supply, inconsistent service quality, poor investments, etc.

Thus, India's urban bus transport systems are overdue for an overhaul. A dual track strategy is needed: improve the narrative of bus systems beyond ridership and simplify investment frameworks to ensure capital flows where it's needed most. This can be done beyond anecdotal accounts to generate robust, data-driven evidence that captures the full spectrum of co-benefits offered by bus systems. A fresh, compelling narrative of quantified benefits of the bus deployment - climate, social, economic and health co-benefits - is urgently needed for India. Without such a comprehensive assessment, policy and funding decisions remain fragmented and reactive, leaving many regions dependent on private vehicles. India's transport sector is undergoing rapid change, with diverse regional trends. A cross-sectional evaluation across diverse geographies and bus systems is necessary. Interestingly, there is a growing portfolio of national and state-level bus funding schemes. A framework for quantifying the benefits of budgetary allocations for bus-based public transportation will help improve investments.

2. Objectives of the BBUS project

In the above context, this project will aim to generate evidence by developing a unique 4-1 co-benefits model covering socio-economic, health, GHG and resilience models to empower local/state/central governments to make data-driven decisions to increase the budgetary allocations and even attract climate funds and private sector investments for bus transport. The project scope encompasses the creation of a "Healthy Cities" Investment Case and a Social Return on Investment (SROI) framework, which will provide a multi-faceted assessment of returns on bus and e-bus investments. This will also help the decision-makers to improve their long-term strategies and demonstrate accountability. This framework is intended for use by the Ministry of Housing and Urban Affairs (MoHUA), NITI Aayog, and state-level transport departments/bus transport agencies to gauge the societal and environmental dividends of transit expenditure accurately.

The key objectives of the project remain:

- Develop an integrated '4-in-1 co-benefits model' to inform policy decisions and support the Monitoring, Reporting, and Verification (MRV) mechanisms of bus systems.
    - M1 - Identify socio-economic co-benefits, especially for women, low-income groups, and other vulnerable road users (VRUs).
    - M2 - Estimate the health and air quality benefits from increased bus usage and reduced reliance on private vehicles.
    - M3 - Quantify the greenhouse gas (GHG) mitigation potential of bus deployment.

- M4 - Assess the resilience and adaptive benefits and impacts of robust bus-based public transport systems on the economy and physical infrastructure in cities.

- Easing access to climate, green funds and other potential avenues for bus and allied infrastructures
- Enhance understanding and belief in bus-based systems: Strategic communications of benefits amongst decision makers and government agencies

3. Approach and Methodology

The methodological approach for this project is rooted in an integrated, mixed-methods framework designed to capture the complex, non-linear impacts of transit systems on society, economy, and the environment. By combining qualitative lived-experience narratives with rigorous quantitative modelling, the research aims to provide a granular view of transport co-benefits. The study will be conducted in two primary groups of cities: Group A - where city buses are operational, and Group B - where city buses will soon be deployed.

3.1. Primary data collection

Data will be collected through a structured questionnaire administered to 6000 respondents across 6 Indian cities. The questionnaire will be self-administered, capturing socio-demographic data, commuter travel behaviour, household expenditure, health and wellbeing indicators, and perceptions of bus service quality. Before questionnaire finalisation, a round of in-depth interviews (IDIs) and focus group discussions (FGDs) will be conducted with diverse user and non-user groups in both Group A and B cities to identify thematic priorities, inform instrument design, and capture lived experiences that quantitative surveys cannot fully represent.

Based on this, the project targets 800-1200 respondents per city, split proportionately between control and experimental groups to enable rigorous impact evaluation. Data collection modalities include:

- Cross-Sectional Surveys: Self-administered questionnaires for Group A participants to assess current operational maturity and perception.
- Longitudinal Assessment: A pre-and-post evaluation for Group B participants. The first assessment occurs before bus deployment, followed by telephonic tracking and a second assessment after operations have stabilised to measure behavioural and health changes.

For the longitudinal assessment in Group B cities, a limited cohort of approximately 450 bus-users (intervention group) and 450 non-users (control group) per city will be tracked over 1–2 years. Telephonic follow-up will be undertaken before the second survey wave to verify continued participation and bus usage status. All data collection will follow informed consent protocols; no sensitive personal data will be collected. Experts from the health industry/academia will be consulted to finalise the health aspect of the study design.

3.2. Geographical coverage

The project aims to be of national relevance by selecting cities with varied levels of bus penetration, operational maturity, and environmental and socio-economic contexts. The project will also aim to achieve geographical diversity with respect to governance architecture. Data will be collected in collaboration with municipal bodies/disaster management offices/state transport understanding (STUs)/planning departments, supported by local research institutions.

Table 1: Tentative list of cities for the project

	Region	City	Population category	Socio-economic status	Extent of supply of PT	Average bus occupancy ratio (2023)	Annual PM2.5 levels (2025)	Climate vulnerability
Group A - Cities with operational bus systems	South	Bengaluru	Mega	Upper-middle income	LOS 2	65%	Non-compliant	Moderate
	North	Jaipur	Metro	Lower-middle income	LOS 4	93%	Non-compliant	High
	East	Bhubaneswar	Large	Lower-middle income	LOS 3	NA	Compliant	High
Group B - Tentative list of cities where buses are soon to be deployed	North	Jodhpur	Medium	Upper-middle income	NA	NA	Non compliant	Very high
	North	Haridwar	Medium	Lower-middle income	NA	NA	Compliant	NA
	East	Gaya	Medium	Low income	NA	NA	Non compliant	Low
	East	Bilaspur	Medium	Lower-middle income	NA	NA	Non compliant	Low
	South	Tirupati	Medium	Lower-middle income	NA	NA	Compliant	High
	South	Puducherry	Medium	Lower-middle income	NA	NA	Compliant	Moderate
	West	Gandhinagar	Medium	Upper-middle income	NA	NA	Non compliant	Low

Authors' compilation
**Population: Below 10 lakh - Medium, 10 to 40 lakh - Large, 40 lakh to 1 Crore - Metro, 1 Crore and above - Mega*
Socio economic status (annual per capita income): Below INR 95000 - Low income, INR 95000 to 370000 - Lower middle income, Above INR 370000 - Upper middle income
Extent of PT supply(Buses per lakh population): Above 60 - LOS 1, 40 to 60 - LOS 2, 20 to 40 - LOS 3, Below 20 - LOS 4
Annual PM2.5 levels: Below 40 µg/m3 - Compliant, Above 40 µg/m3 - Non-compliant
Climate vulnerability (exposure to events - floods, droughts, cyclones):

Group A: Cities with bus systems

- **Bengaluru:** A megacity in South India featuring a **well-maintained fleet** and a broad user spectrum. It is an essential site for studying transit-oriented resilience amid escalating climate vulnerabilities, such as **seasonal flooding** and urban heat.
- **Jaipur:** One of the **fastest-growing cities in Northern India**, with a fleet of buses operational for more than 10 years and new bus inductions scheduled in the coming year. It also represents a

tourist mix in its service coverage. Moreover, the city witnesses **heavy air pollution** during the winter season.

- **Bhubaneswar:** An emerging city with a modern, award-winning model of how a mid-sized city can leapfrog to a **tech-enabled, inclusive transit system**. It also represents a site with high climate vulnerability, especially **heavy rains and cyclones**.

Group B: Cities with soon to be operational bus systems
A tentative list has been prepared, and soon, based on discussions with the Ministry of Housing and Urban Affairs (MoHUA), the cities will be finalised. This is necessary in terms of understanding the timelines of e-bus deployment as part of the PM e-Bus Sewa scheme of MoHUA.

3.3. Modelling approach

- **Model 1 — Socio-Economic Co-Benefits:** M1 will assess differential impacts on commuters, including vulnerable road users (VRUs) - women, children, senior citizens, persons with disabilities, and economically weaker sections. Multivariate regression and disaggregated analysis will identify shifts in household expenditure, educational access, economic participation, and time savings attributable to improved bus access.
- **Model 2 — Health Co-Benefits:** M2 will use a mixed cross-sectional and longitudinal design to quantify health impacts of bus usage, including reductions in air pollution exposure, road traffic injuries, mental stress, and gains from increased active travel. Self-reported health survey forms validated by local medical practitioners will be used for the cross-sectional component. For the longitudinal component in Group B cities, standardised health outcome indicators will be mapped over the study period. Generalised linear models and multivariate analysis will be employed to control for confounding variables across income, gender, education, and regional strata.
- **Model 3 — GHG Quantification:** M3 will construct a bottom-up GHG emissions model aligned with ITF-OECD and UNFCCC methodologies, customised to India's mixed-traffic conditions. Empirical modal shift data from travel diaries, Origin-Destination (OD) surveys, and automated vehicle counts will feed into calculations of avoided vehicle kilometres travelled (VKT). Dynamic emission factors will be derived from city-specific fuel mix data, grid emission intensities, and e-bus depot charging patterns. Scenario analysis will project GHG savings under current policies, renewable integration, aligning with IEA analytical frameworks.
- **Model 4 — Resilience and Climate Adaptation:** M4 will evaluate the resilience of bus systems during natural and man-made hazards, including urban floods, cyclones and heat waves. Three components will be integrated: (i) disaster response modelling using GIS-based routing optimisation and evacuation analysis; (ii) economic resilience assessment comparing cumulative losses under restricted mobility for bus-users versus private vehicle users; and (iii) self-resilience of bus infrastructure, including comparative downtime, asset losses, and restart costs across transport modes. Data will be sourced from municipal disaster management offices, hazard maps, demographic profiles, and shelter location databases.

4. Outputs and outcomes

4.1. Develop a tool for the 4-1 co-benefit model

The project will aim to create an online, open-access Pre-Investment Appraisal Tool based on the 4-in-1 model. This tool would allow a city transport official or state government planner to input basic parameters — fleet size, city population, current modal share, grid emission factor — and receive a standardised output showing benefits such as *(indicative list)*:

- Projected GHG savings (tonnes CO_2e/year) with scenario analysis
- Estimated health co-benefits (DALYs averted, hospitalisation costs saved)
- SROI ratio (₹ societal return per ₹ invested)
- Eligible climate finance instruments with indicative funding amounts
- MRV indicator dashboard aligned with India's NDC/state climate targets/SDGs

This tool directly addresses the capacity gap in smaller cities and STUs, where the absence of technical expertise is a primary barrier to accessing climate funds. It also creates a standardised pre-feasibility document that is already formatted to meet the project preparation requirements of GCF, ADB, and World Bank, dramatically reducing the transaction cost of applying for multilateral financing.

4.2. Healthy cities – SROI on bus-based public transport

The models created using this project will serve as a data-driven de-risking toolkit for bus transport agencies. The investment case will be built on the monetisation of the co-benefits established in the project, aligning with financial metrics such as IRR, NPV, cost per km, etc. Each model output will give a distinct lens to the returns from bus transport systems and their linkage to better investments (Table 2).

Table 2: Project's 4 co-benefit models and their respective investment access instruments

Model	Types of co-benefits	Monetisation metrics	Target investment channels to unlock	Key agencies/ instruments
M1 - Socio-economic	Household savings, time savings, female workforce participation	• Productivity value increased (₹/hr) • Household transport expenditure reduction (₹/month per commuter) • Female labour force participation uplift monetised via earnings gain • GBA+ disaggregated benefit accounting (women, PwDs, elderly, EWS)	• CSR funding • MGNREGS-aligned state spending • Gender lens investment • Social protection budgets	ADB SIDBI State Finance Commissions MoHUA CITIIS
M2 - Health	Avoided DALYs, hospitalisation costs, road	• WHO Value of Statistical Life (VSL) adapted to India's	• Health-transport convergence VGF • State health	MoHFW / NHM WHO India

	fatalities, pollution exposure	income levels ● DALY-based cost-of-illness valuation (₹ per DALY averted) ● Avoided public health expenditure (hospitalisation, treatment costs) ● Health Impact Assessment (HIA) → ₹ crore saved/year	departments co-finance ● National Health Mission budgets ● Results-based financing (RBF)	
M3 - GHG mitigation	Avoided VKT, modal shift, reduced CO_2e, e-bus grid emissions	● Social cost of carbon valuation ● India's shadow carbon price applied to avoided emissions ● Scenario analysis: current policy / SDS / 100% renewable grid ● Dynamic emission factors (city-specific grid mix, charging patterns) ● Pre-certified carbon credit methodology (ICAT / Article 6.2)	● Green Climate Fund (GCF) ● Climate Investment Funds (CIF) ● Green bonds / SLBs ● Article 6 bilateral carbon markets ● MDB concessional finance	GCF / CIF ADB AIIB World Bank KfW SEBI Green Bond Framework UKAID
M4 - Resilience and Adaptation	Avoided economic downtime, disaster routing, and infrastructure self-resilience	● Avoided economic downtime value (₹ crore/event vs. rail/private modes) ● GIS-based disaster routing — evacuation time savings monetised ● Comparative infrastructure damage and restart costs across modes ● Economic resilience: captive commuter loss avoided during disruption	● NDMA ● NAFCC ● ADB ● World Bank ● GCF Adaptation Window	

Source: Authors' compilation

Based on the 4 models, a layered SROI Framework will be developed as the core investment instrument, designed specifically for multiple investor audiences. The approach should produce three SROI lenses from the same underlying dataset:

- **SROI Lens 1 — For Government (Central, State and Municipal):** Present the ratio of total societal co-benefits (health savings + productivity gains + GHG value + resilience value) to public expenditure per bus per year. This directly addresses the "value for public money" question that government agencies seek. For instance, the PM-eBus Sewa scheme already provides ₹24/km operational VGF — the SROI should show what societal return that ₹24/km is generating.
- **SROI Lens 2 — For Climate and Development Finance Institutions:** Structure the co-benefits in alignment with the Green Climate Fund's results framework (mitigation outcomes, adaptation outcomes, sustainable development co-benefits) and ADB's Climate Change Operational Framework. This means expressing outputs as: tonnes CO_2e avoided per year, number of vulnerable people benefited, DALYs averted per 10,000 population, and economic resilience value in ₹ crore per disaster event. These metrics directly match the project appraisal criteria of GCF, ADB, World Bank, and bilateral funds (UKAID, GIZ, AFD).
- **SROI Lens 3 — For Private Sector and Capital Markets:** Develop a blended finance prospectus that uses the co-benefit data to de-risk private investment. The argument is: if governments are capturing ₹X in health and productivity savings per bus, a portion of those savings can be structured as a payment security mechanism or revenue guarantee for private operators. This is precisely the model being explored under the Payment Security Mechanism (PSM) for PM-eBus Sewa. The 4-in-1 model provides the empirical underpinning to size and justify that guarantee.